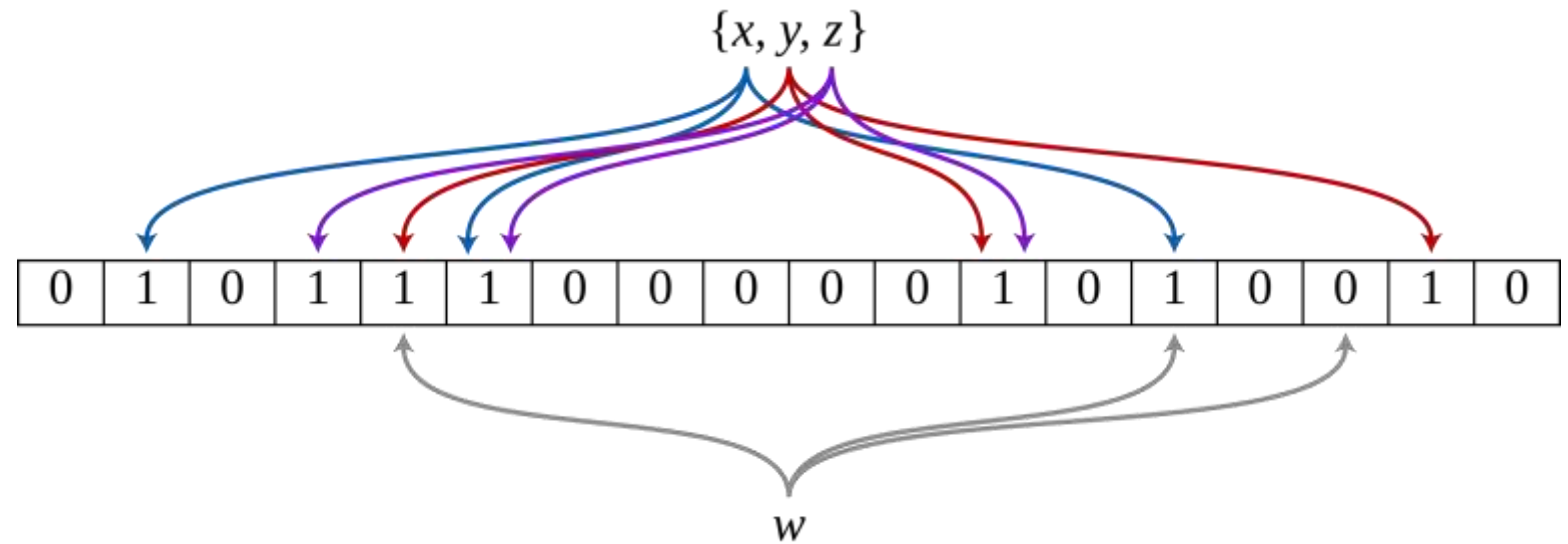

FAST LEARNING MANAGEMENT WITH BLOOM FILTERS

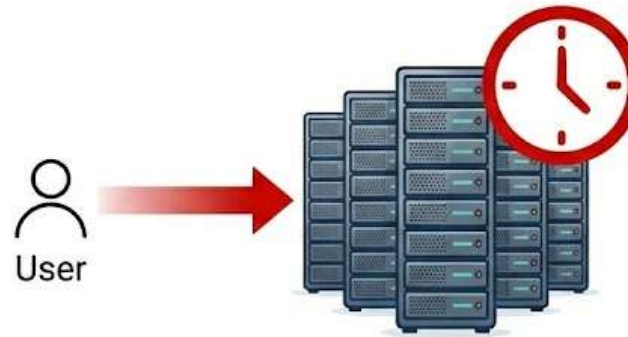


Arda Durmaz
23070001009

THE PROBLEM: DATABASE BOTTLENECK

Before and After

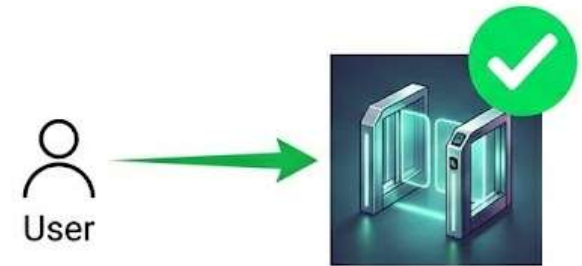
Standard LMS Query



Full Database Scan

Slow Disk I/O &
Expensive Searches

The Goal

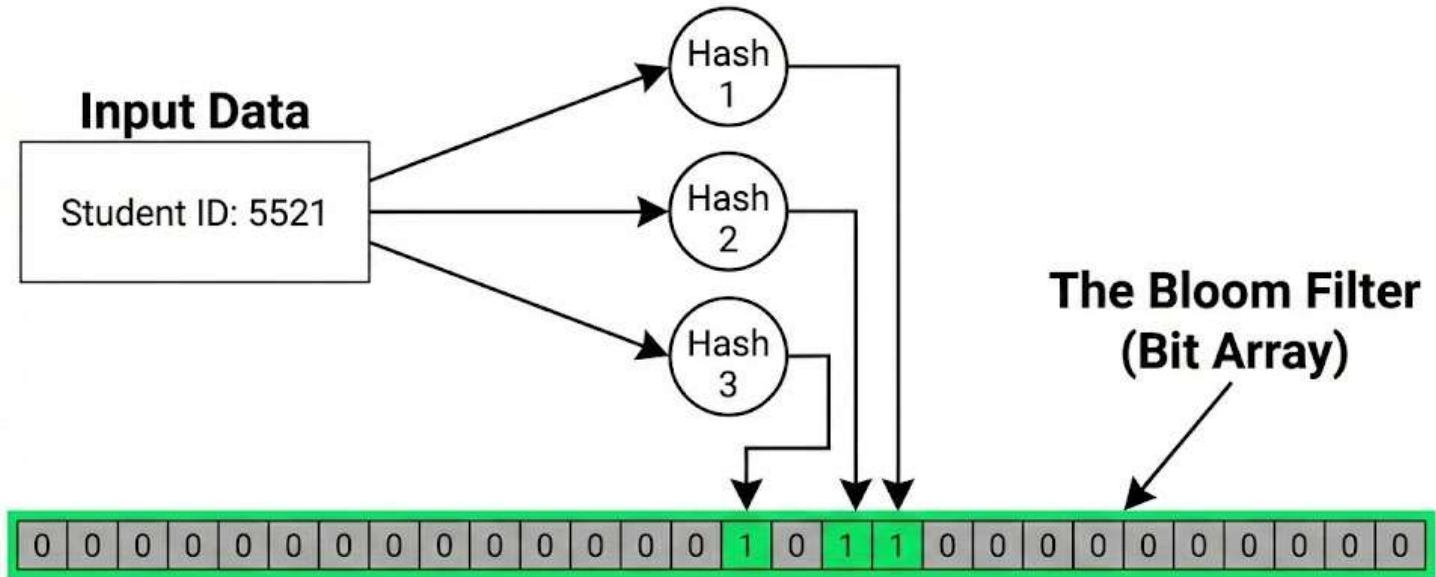


Gatekeeper

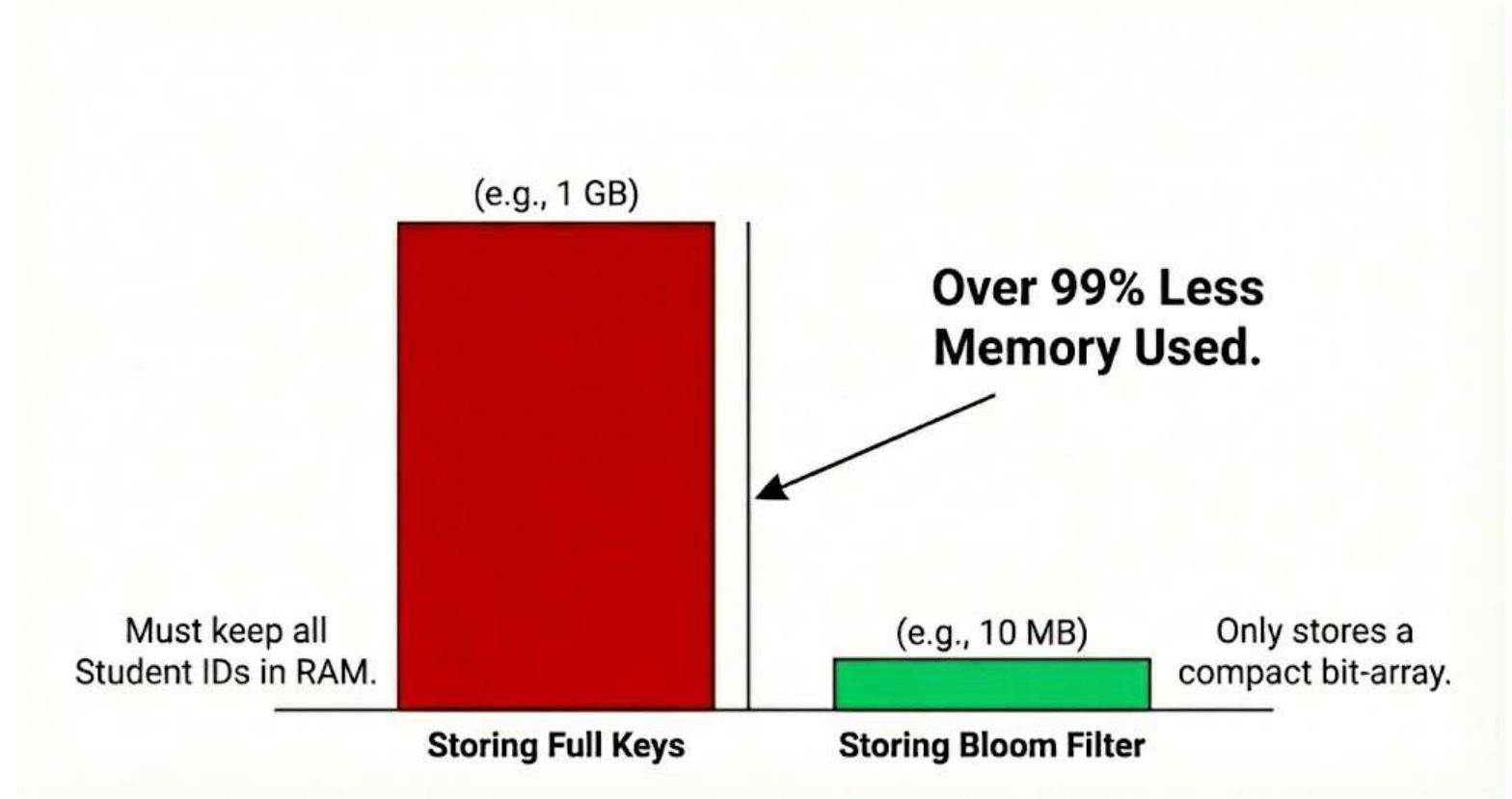
Instant Answer:
"Does this record exist?"

THE SOLUTION:BLO OM FILTER

Bloom Filter Works



COMPARIS ON



FALSE POSITIVE PROBABILITY

False Positive Probability Formula

$$p \approx (1 - e^{-kn/m})^k$$

p = False positive probability

k = Number of hash functions

n = Number of elements inserted

m = Total bits in the filter



THANK YOU

- REFERENCES:
- [MurmurHash - Wikipedia](#)
- [Bloom filter - Wikipedia](#)
- [Jenkins hash function - Wikipedia](#)